

# Euclid's Elements — Book I

## Proposition 31

CGJteamLab

*Through a given point to draw a straight line parallel to a given straight line.*

### 1 Introduction

Proposition I.31 is a construction theorem:

given a straight line  $AB$  and a point  $P \notin AB$ ,

construct a straight line through  $P$  parallel to  $AB$ .

At first sight the proposition belongs naturally to the Euclidean theory of parallel lines developed in I.29–I.30. Its logical status is different, however. Proposition I.31 does not require Hilbert's Group IV. It is an existence theorem in neutral geometry.

The construction uses an arbitrary transversal through the given point, copies an angle by Hilbert's angle-construction axiom, and then applies the neutral alternate-angle criterion corresponding to Euclid I.27.

Thus the essential mechanism is

angle construction  $\longrightarrow$  equal alternate angles  $\longrightarrow$  parallel lines.

No numerical angle measure and no construction of a right angle are needed.

### 2 Classical Configuration

Let  $AB$  be the given straight line and let  $P$  be a point outside it.

The goal is to construct a point  $Q \neq P$  such that

$$AB \parallel PQ.$$

Choose the line  $AP$  as a transversal. Then choose a point  $E$  with

$$A - E - P.$$

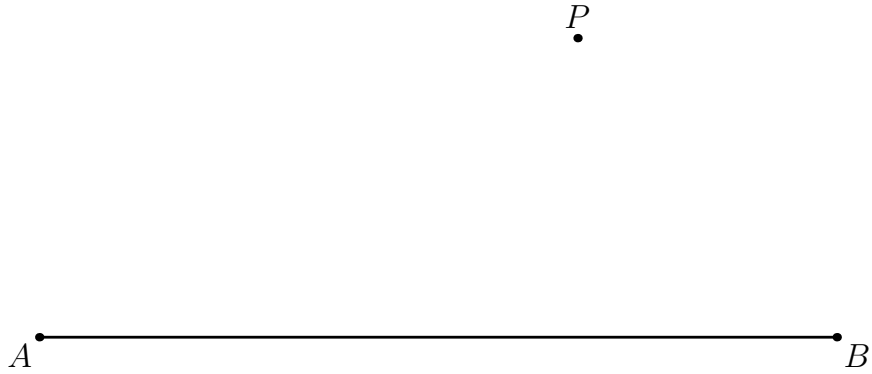


Figure 1: The initial data for Proposition I.31: the given straight line  $AB$  and the external point  $P$ .

The source angle is

$$\angle EAB.$$

At  $P$ , with  $PE$  as the first ray, construct a ray  $PQ$  on the appropriate side of the transversal such that

$$\angle EAB \cong \angle EPQ.$$

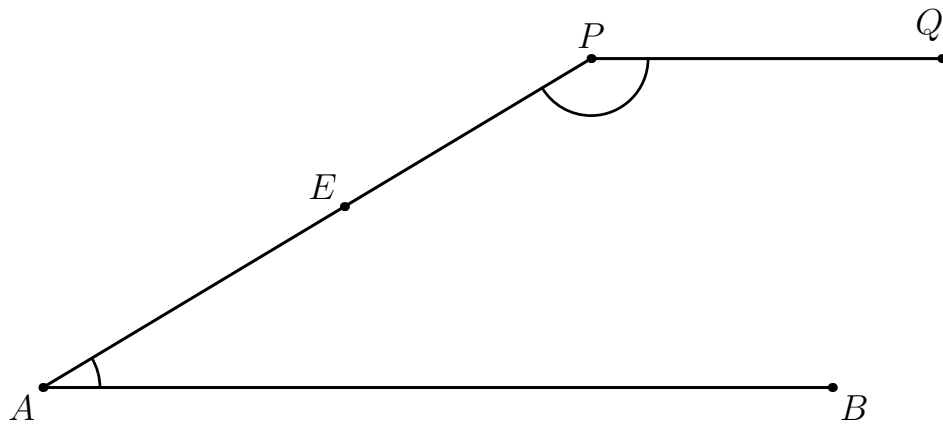


Figure 2: The transversal  $AP$ , the point  $E$  with  $A - E - P$ , and the copied angle at  $P$ .

The two congruent angles are alternate angles for the lines  $AB$  and  $PQ$ . Proposition I.27 therefore gives

$$AB \parallel PQ.$$

### 3 Why No Right Angle Is Needed

The construction does not proceed through perpendicular lines.

The angle

$$\angle EAB$$

is completely arbitrary. Its numerical size is irrelevant. Hilbert's angle-construction principle allows this angle to be copied at  $P$ , with the new ray placed on a prescribed side of the transversal.

Thus the direction of the new line is transferred synthetically by angle congruence:

$$\angle EAB \cong \angle EPQ.$$

The parallelism then follows from the converse alternate-angle theorem, that is, Proposition I.27.

This is conceptually important. The construction of a parallel does not require a prior theory of perpendiculars or a notion of  $90^\circ$ . It requires only the ability to reproduce a given angle and control its position relative to the transversal.

## 4 The Hidden Configuration

A classical diagram makes the side relation almost invisible. In the formal proof it must be explicit.

The equality

$$\angle EAB \cong \angle EPQ$$

alone is not sufficient. The point  $Q$  must be chosen so that  $B$  and  $Q$  lie on opposite sides of the transversal  $AP$ .

The formal construction therefore contains three distinct ingredients:

$$A - E - P,$$

which fixes the directions of the transversal rays;

$$B \notin AP,$$

which guarantees that the source angle is nondegenerate; and

$$B, Q \text{ on opposite sides of } AP,$$

which identifies the copied angles as genuine alternate angles.

This is exactly the type of configuration information that is compressed into an ordinary Euclidean picture but becomes explicit in a Hilbert-style formalization.

## 5 Hilbert Reconstruction

The substantial geometric content is isolated in the reusable theorem

hilbert\_parallel\_through\_point\_exists

with the formal statement

```
theorem hilbert_parallel_through_point_exists
  [HilbertCongruence Geo]
  (A B P : Geo.Point)
  (hAB : A != B)
  (hABP : not Collinear Geo A B P) :
  exists Q : Geo.Point,
    P != Q /\
    Geo.Parallel A B P Q
```

The proof first constructs the incidence line through  $A$  and  $B$ , and then the transversal through  $A$  and  $P$ .

Next it chooses  $E$  with

$$A - E - P.$$

The non-collinearity assumption implies that  $B$  lies off the transversal, so the angle  $\angle EAB$  is genuine.

A point  $S$  is then chosen on the side of the transversal opposite to  $B$ . Hilbert's angle-construction principle constructs  $Q$  on the side of  $S$  such that

$$\angle EAB \cong \angle EPQ.$$

The side relation is transported from  $S$  to  $Q$ , yielding the required opposite-side configuration for  $B$  and  $Q$ .

Finally,

hilbert\_parallel\_of\_alternate\_angles\_oppositeSide\_lines

applies the neutral alternate-angle criterion and returns

$$AB \parallel PQ.$$

## 6 Lean Formalization

The Book I theorem is

```
theorem euclid_proposition_31
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B P : Geo.Point)
```

```

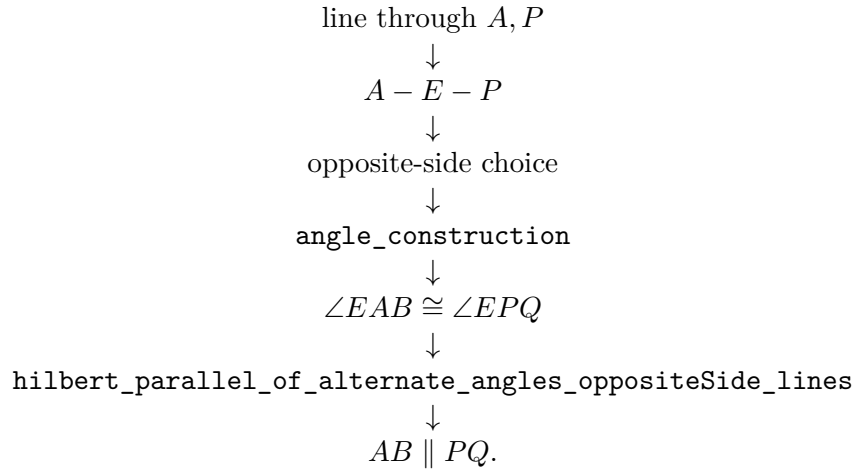
(hAB : A != B)
(hABP : not Collinear Geo A B P) :
exists Q : Geo.Point,
  P != Q /\
    Geo.Parallel A B P Q := by

exact
  hilbert_parallel_through_point_exists
    Geo
    A B P
    hAB
    hABP

```

Thus the Book I file is only a wrapper. The reusable geometric theorem belongs to the Hilbert interface, while the Proposition file records the corresponding result in Euclid's sequence.

The dependency graph of the actual Lean proof is



## 7 Neutral Existence Versus Euclidean Uniqueness

Proposition I.31 establishes the existence of a parallel through an external point.

It does not establish uniqueness.

This distinction separates I.31 from the immediately preceding Euclidean parallel propositions. In the Hilbert reconstruction, uniqueness is the additional content of Group IV:

through a point outside a line there is at most one parallel.

Hence

I.31: existence

belongs to neutral geometry, while

Group IV: uniqueness

is the Euclidean addition.

This also explains a useful structural relation with Proposition I.29. Once a parallel has been constructed by the mechanism of I.31, a uniqueness principle such as Playfair’s axiom can identify that constructed line with an already given parallel through the same point. This observation belongs to the logical analysis surrounding I.29; it is not an additional step in the proof of I.31 itself.

## 8 Final Configuration

The construction produces a point  $Q \neq P$  satisfying

$$AB \parallel PQ.$$

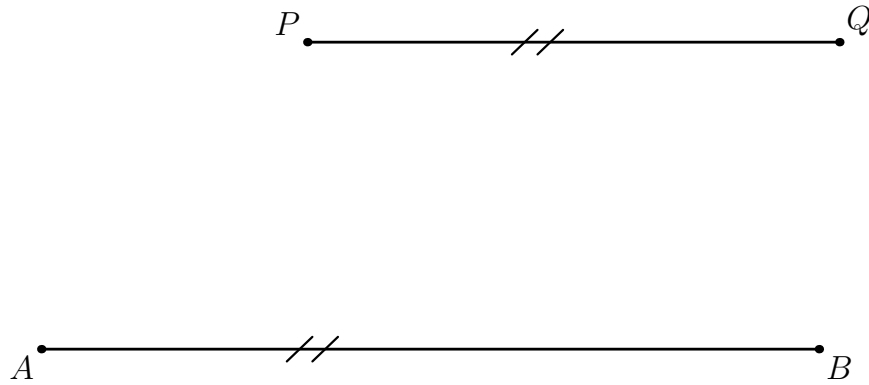


Figure 3: The final configuration of Proposition I.31: the constructed line through  $P$  is parallel to the given straight line  $AB$ .

## 9 Conclusion

Proposition I.31 is a particularly clear example of the difference between existence and uniqueness in the theory of parallels.

Its proof requires no Euclidean parallel axiom. A transversal is chosen, an arbitrary angle is copied at the external point, the side configuration is controlled explicitly, and Proposition I.27 proves that the resulting line is parallel to the given line.

The Hilbert reconstruction therefore isolates the reusable theorem

`hilbert_parallel_through_point_exists`

as a neutral construction result.

The Book I theorem

`euclid_proposition_31`

is then a thin wrapper around this more general piece of the developed Hilbert theory.